Emergent Perception of Gridletters

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Abstract

This paper introduces a cognitive model of letter perception based on the principles of emergent computation and high-level perception. The model goes beyond simple categorization by parsing a letterform into its constituent parts. Such parts are formed under the top-down influence of letter concepts. The model currently recognizes letters in many diverse typefaces and will be incorporated into a larger model of creativity — Letter Spirit.

1 Letter recognition at the conceptual level

This paper briefly describes a cognitive model of letter perception that is being used as part of the Letter Spirit project.[1] The model is based on the tenets of high-level perception, as described in [Chalmers et al., 1992], in which concepts dynamically provide top-down influence on the formation of perceptual structures during the process of categorization. The resulting perceptual structures, which emerge from the stochastic activities of a large number of tiny processing agents, are well-suited for further analogical processing [Mitchell, 1993]. These perceptual structures include important information about the conceptual makeup of a letterform (as opposed to a simple categorization like "is an 'a'"). This "conceptual parsing" turns out to be critical to the *design* of stylistically similar letterforms — the ultimate aim of Letter Spirit.

Most optical character recognition (OCR) models take cognitively unrealistic approaches to letter recognition. These approaches share the quality of being shape-based and relying heavily on low-level, syntactic features [Mantas, 1986]. In contrast, our psychological studies of human letter recognition indicate that processing occurs at a much higher cognitive level. We hypothesize that letter concepts are composed of a set of conceptual pieces that we call *roles* [McGraw et al., 1994a]. The role hypothesis, briefly described in section 3, is supported by strong psychological evidence [McGraw et al., 1994b]. This paper presents a model of letter perception based on the role hypothesis.[2]

Why focus on cognitive plausibility of recognition models instead of engineering efficiency? Because human recognition of letterforms (especially highly-stylized ones) is still far superior to that of machines. We believe that a human-like approach will significantly enhance the computer's ability to correctly recognize a wide variety of non-standard letterforms. We are driven by the need for a powerful recognizer to use in the Letter Spirit model of typeface design.

2 Letter Spirit, gridletters and gridfonts

The Letter Spirit project is an attempt to model central aspects of human high-level perception and creativity on a computer, focusing on the creative act of artistic letter-design. The aim is to model the process of rendering the 26 lowercase letters of the roman alphabet in many different, internally coherent styles. Two important and orthogonal aspects of letterforms are basic to the project: the *categorical sameness* possessed by instances of a single letter in various styles (*e.g.,* the letter 'a' in Baskerville, Palatino, and Helvetica) and the *stylistic sameness* possessed by instances of various letters in a single style (*e.g.,* the letters 'a', 'b', and 'c' in Baskerville). Figure 1 shows the relationship of these two ideas.

Starting with one or more seed letters representing the beginnings of a style, the program will attempt to create the rest of the alphabet in such a way that all 26 letters share the same style, or *spirit*. To avoid the need for modeling low-level vision and to focus attention on the deeper aspects of letter design, we

[1] For information about the on-going Letter Spirit project see [Hofstadter and the members of FARG, 1995] and [McGraw and Hofstadter, 1993] which are available on the World Wide Web through URL `http:\\www.cogsci.indiana.edu`.

[2] The model described below can be contrasted with the simplistic Netrec and Dumrec models introduced in [McGraw and Drasin, 1993] and criticized in [McGraw et al., 1994a].

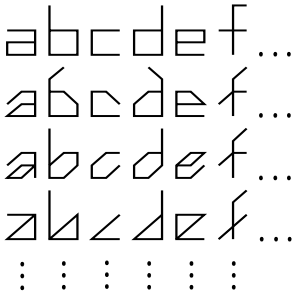


Figure 1: Items in any column have *letter* in common. Items in any row have *spirit* in common.

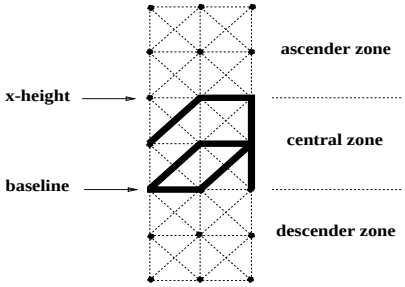


Figure 2: The Letter Spirit grid, with quanta instantiating one of many possible 'a's turned on.

developed an idealized micro-domain. Letterforms are restricted to short line segments on a fixed grid of 21 points arranged in a 3×7 array. Legal line segments, called *quanta*, connect a point to any of its nearest neighbors. There are 56 possible quanta, as shown in Figure 2. This restriction allows much of low-level vision to be bypassed and forces concentration on higher-level cognitive processing, particularly the abstract and context-dependent character of concepts.

Because quanta are either on or off, decisions on the grid are coarse. Surprisingly, the variety among letters of a given category is still huge — hundreds of versions of each letter and 600 full gridfonts have been designed by humans. Almost paradoxically, the domain's limitations engender this diversity. As there are no surface features to manipulate at a fine-grained level, one ends up playing at the boundaries of the 26 categories. Consequently, many gridfonts are wild, sometimes having angular, blocky, spiky, sparse, or otherwise bizarre letters. This makes the task of recognizing gridletters hard, for both humans and machines.

3 Letter concepts and the notion of roles

Figure 3 is a graphical representation of three common ways one can break the letter 'b' into conceptual pieces. We call the conceptual pieces *roles.* Roles and relationships between roles (shown as black dots in the figure) make up the internal structure of a letter category and together define a particular *conceptualization* of a letter. Category membership at the whole-letter level is partially determined by category membership at the lower level of roles. (We say "partially" because the interaction between roles also matters. For example, a graphic shape might have a strong exemplar of *post* to the left of a strong exemplar of *loop* and yet the way they interact might still make them look more like 'lo' than 'b'.

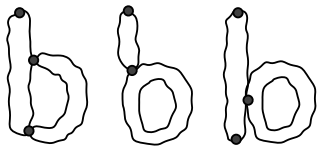


Figure 3: Three common conceptualizations of the letter 'b'. Each conceptualization has two roles.

Graphic letterforms are comprised of *parts* that fill a letter-conceptualization's roles. During perception, representations of such parts are formed under top-down pressure from roles and are sensitive to context. As stated in [Palmer, 1978, p.96], "components [of parts] enter into relationships with other components, resulting in larger structural units whose importance supersedes that of its constituents." We hold that most of the "importance" attributed to the emerging parts stems directly from their role-filling ability. In other words, the way in which a part fills a role directly determines its "goodness."

Sometimes norms associated with roles must be *violated* in order to accommodate a given letterform's parts. In this sense, the roles are "plastic". This plasticity is a result of allowing certain slots in the mental representation of a role to be filled by any of a range of closely related part-descriptions. The plasticity of norms associated with a role is context-sensitive and varies according to conceptual pressures brought to

bear by the situation. Some of the norms making up the representation of a role are more malleable than others. We argue that the non-brittle nature of human letter perception is due to plasticity at the level of roles.

4 Letter Spirit's Examiner

The gridletter recognizer in Letter Spirit is called the Examiner. All perceptual and creative processes in the Examiner are emergent, in the sense that they result from the actions of a large number of independent *codelets* — computational micro-agents that create, examine, and modify structures representing parts, roles, letters, stylistic traits, and so forth — and that they do so in (simulated) parallel. Newly created codelets wait to be run in a structure called the *Coderack*, which can be thought of as a stochastic waiting room. In contrast to a standard operating-systems queue, where processes wait before being deterministically given their slice of CPU time, the Coderack features stochastic selection of actions. To each codelet is attached an *urgency* value — a number that determines its probability of being chosen next. Urgency values are based on how well a codelet's possible effect coheres with structures already built.

Actions of every sort — gluing, labeling, scanning, matching, adjusting, regrouping, destroying, and so on — are carried out by codelets. The effect of each codelet considered by itself is very slight; however, as many codelets run, their independent effects build upon one another into a coherent collective behavior.

A useful image is that of a large structure (like a bridge) being built by a colony of hundreds of ants or termites. The ants work semi-independently but cooperatively. Codelets correspond to the ants in this metaphor, and perceptual structures to the bridge. So perceptual structures develop nondeterministically but not haphazardly. From a large number of very small probabilistic decisions and actions, coherent perceptions are built up.

Over a long period of time, *processes* are interleaved in a manner reminiscent of time-sharing. (A process consists of many codelets, which *ex post facto*, can be seen to have been acting in concert.) One notable difference between this and conventional time-sharing is that the biased nondeterministic selection of codelets amounts to having different processes run at different speeds. The speeds themselves can be regulated over time, by varying the urgencies of the codelets involved, to favor more-promising directions over less-promising ones. Since codelets have very small effects, it is never critical that any particular codelet get selected. What does matter is that certain broad-stroked courses of action as a whole run faster than others. Probabilistic selection based on urgencies allows this to happen.

4.1 Processing in the Examiner

The Examiner operates roughly as follows, although the following outline may give the impression that processing is more serial and well-ordered than it really is. In reality, processing occurs in a more parallel manner, with various actions described below often proceeding concurrently.

- Quanta are probabilistically bonded together with different amounts of "glue". (For example, more glue tends to be deposited at straight junctions than at angles.) The glue-depositing agents execute completely bottom-up syntactic operations.
- When enough glue has been deposited, the glued shape is metaphorically "shaken". This amounts to probabilistically breaking the glued shape into parts at weak joints, resulting in a set of chunks (usually made up of between two and four quanta) called *parts*. Since no concepts have been invoked up to this point, parts are still purely syntactic entities.
- Each part is scanned by many agents that probabilistically attach *syntactic labels* to the part. Labels denote very simple properties of parts like curviness, length, width, location, and so on, and in no way involve the set of categories into which these stimuli will eventually be channeled.
- The presence of a particular label on a given *part* serves as a cue that tends to lightly activate one or more *roles* with which the label is associated (*i.e.,* roles of which the given label is at least somewhat indicative). For instance, the labels "left", "ascender", and "straight" would tend to activate the "left-post" roles of 'b'. It is important to understand that even a label such as "straight" is probabilistic, in the sense that a not totally-straight part *might* get that label, with probability diminishing with its non-straightness.

- The various light activations sum up to a *total* activation-level for each role. Any role with sufficient activation will attempt to "mate" with the specific part whose labels activated it (here is where syntax and semantics meet). Roles compete with each other for a given part. Usually the role that best matches a part will win a fight, but once again, role-matching is probabilistic. A complementary way of describing the process is to say a given part competes for the attention of various roles.
- As roles and parts attempt to couple, a given part may need to be slightly altered in order to be a good mate for a given role. A quantum or two may need to be stolen from one or more neighboring parts to make the part in question more attractive to a role. Likewise, small pieces that seem to make a part ugly in the eyes of possible role-mates may need to be detached. (Note that only the parts are undergoing regrouping; the underlying quanta defining the letterform remain the same.) The resulting structures composed of quanta are now results of the combined influence of bottom-up and top-down processing. As such, these parts are no longer totally syntactic entities, and we call them "(semantically) adjusted parts."
- Roles compete for parts throughout the letter, adjusting the parts as they go. When this adjustment-and-mating phase is over, there should be a fairly strong match-up between roles and semantically-adjusted parts.
- Each instantiated role will have a few tags attached, stating *how well* the given part instantiates the role. This information will focus on how the part deviates from various norms associated with the role. For example, a 't' whose spine is too tall or bent over at the top, or whose crossbar is too short, too high, or tilted will have tags stating such things.
- Each filled role begins to alert one or more *wholes* (*i.e.,* full letter-concepts such as 'a') for which it provides evidence, in the sense of fitting a particular conceptualization of that letter. Role/whole coupling is analogous to part/role coupling, only it occurs at a higher (more semantic) level. Particular letter-conceptualizations will be activated according to how strongly their component roles are realized in the actual grid letter.
- Different wholes thus become activated to different extents.
- If there is a clear leading contender among the wholes, it is deemed the winner. If there is a close race between several, the letterform is deemed ambiguous and therefore unacceptable. In a borderline case between clear-winner and close-race, a probabilistic decision is made that chooses between the two courses of action.

4.2 A simple example

Figure 4 shows that the Examiner was able to recognize a particular non-standard 'b'. This screendump was made at the end of a run.[3] Notice that the roles *left-post* and *left-bowl* are highly active. So is the correct category *plato-b*. The two ultimately successful parts were formed dynamically under top-down pressure from the existing roles that the Examiner knows about. The syntactic labels attached to each of the letterform's two parts are shown to the right of the letterform itself. It is these labels — probabilistically attached during the run — that activated the appropriate roles. Glyphs representing the active roles are superimposed on the parts during the course of a run with highly active roles drawn using heavier lines. All processing is dynamic — with parts, labels, and activations adjusting and changing throughout a run.

5 Conclusions

The Examiner has been implemented and is now being "tuned" and tested. Preliminary results are promising, with the program exhibiting the ability to recognize (i.e., categorize *and* parse into roles) many diverse gridletters taken from all of the 26 letter categories. When the program is complete, its performance will be compared and contrasted to human performance based on letter recognition data reported in [McGraw et al., 1994b]. Completion of the Examiner represents an important milestone on the way to the full implementation of Letter Spirit.

[3] The Examiner's display is in color, which makes understanding the information easier.

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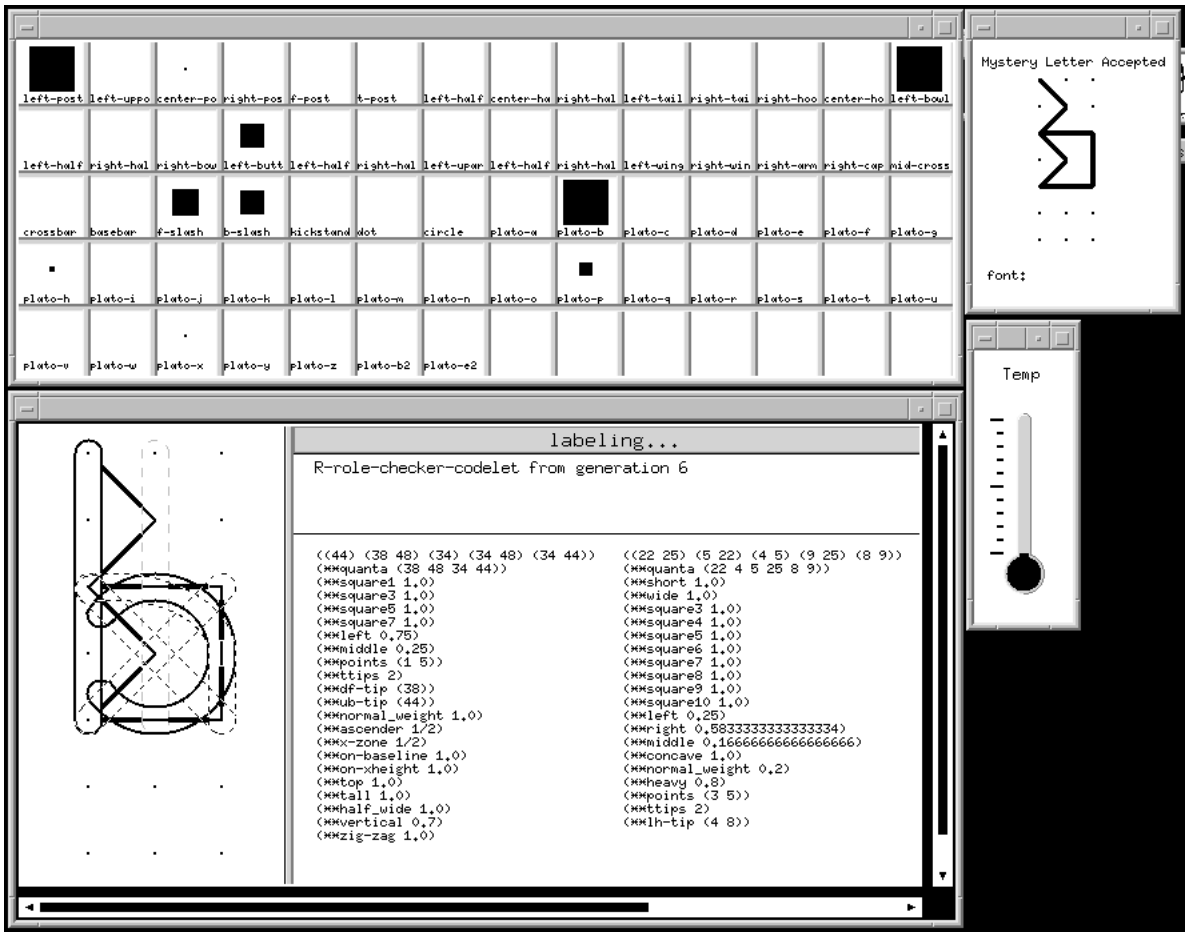


Figure 4: Categorization and "parsing" into roles of a non-standard 'b'. Activations of roles and role-sets are shown in the top window. The workspace where parts are formed, labels are attached, and perceptual structure is built is shown in the bottom window. An unparsed version of the gridletter was entered in the "Mystery letter" window before processing began.